Maxwell Tsai

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EDUCATION

Boston University – College of Engineering Expected May 2027
Bachelor of Science in Electrical Engineering GPA: 3.96/4.00

Coursework: Circuits and Electronics, Power Electronics (Spring 2027), Electric Energy Systems, Electromagnetic Systems, Control Systems, Signals and Systems, Logic Design

EXPERIENCE

Intel Hillsboro, OR
Data Science and Analytics Intern May 2026 – Aug 2026

- Cut wafer data retrieval from 2+ weeks to 2 hours (~100 measurements, ~200 wafers) by building a data-pulling tool in JMP Scripting Language with device and automation teams; adopted by the team after the internship
- Traced errors in a ridge regression model predicting in-line electrical test values to outdated metrology revisions in the training data; the tool keeps only the latest revision per measurement per wafer

Analog Product Development & Test Engineer Intern May 2025 – Aug 2025

- Architected an object-oriented Python API abstracting 4 lab instruments, including a Bit Error Rate Tester, as generalized scopes, clocks, and sources, letting engineers test remotely with common functions instead of instrument-specific commands
- Automated IEEE 488.2 SCPI communication over GPIB (PyVISA) to configure instruments and transfer measurement data
- Refactored 7,000+ lines of Python to PEP 8 standards and wrote ~100 unit tests covering API functions for each instrument

Boston University Dept. of Electrical and Computer Engineering Boston, MA
Teaching Assistant – Logic Design Jan 2026 – May 2026

- Led Verilog FPGA lab sections in Vivado and debugged student designs; graded homework and exams for 80+ students

SemeterLab at Boston University Boston, MA
Undergraduate Research Assistant Sep 2025 – Dec 2025

- Developed an Android app logging Global Navigation Satellite System (GNSS) measurements for real-time ionospheric monitoring, tracking 15+ satellites and exporting carrier-to-noise density (C/N0) data

PROJECTS

BusBot: Autonomous Dish-Collection Robot (Power and Sensor Lead) May 2026 – Present

- Sized a 12 V, 18 Ah SLA battery for a 2.5 hr runtime requirement from a component-level power budget (~35 W average draw at 36 kg loaded weight)
- Designed the 12 V power front end with a 20 A fuse, E-stop and fused branch circuits, using an INA226 over I2C for low-battery detection and motor current measurement under load
- Defined sensor coverage for LiDAR navigation, adding ultrasonic sensing to cover the 2D 276º LiDAR's rear blind spot within a 0.2 m obstacle buffer
- Wrote ESP32-S3 firmware (ESP-IDF, C) for obstacle stopping with PWM motor control and quadrature encoder feedback: brakes at 40 cm, resumes past 45 cm, and stops on sensor failure
- Validated on a motor bench rig: ±2 cm range accuracy, brake command under 1 ms after detection, 5/5 stop/restart trials

Autonomous Laundry Folder Jan 2026 – May 2026

- Built the power system for a 5-person team's SCARA-inspired 2R1P manipulator: a 12 V supply driving an electromagnet end effector, with buck converters for the Arduino and two 25 kg·cm linkage servos
- Derived the manipulator's forward kinematics by hand and debugged Arduino joint-angle control against a teammate's inverse kinematics solution, reaching correct joint angles at each step of a 10-step trajectory

Bedside Beacon: Point-to-Point Communication Device Sep 2025 – Dec 2025

- Built servo-steered laser transmitter and photodiode receiver circuits with ASCII pulse encoding; 29 of 35 transmissions (83%) succeeded at 6–15 ft, up to 3x the 5 ft design spec

SKILLS

Design: KiCAD, MATLAB, Onshape (CAD)
Programming: Python (pandas, pytest, PyVISA), C++ (Arduino), C (ESP-IDF), Verilog, JMP Scripting Language
Lab: Oscilloscope, Bit Error Rate Tester, Waveform Generator, DC Power Supply, DMM, GPIB/SCPI
Tools: Git, Linux, Xilinx Vivado, JMP

LEADERSHIP

Theta Tau (New Member Education, Professional Development and Brotherhood Chair): ran resume and interview workshops and coached 20+ members 1-on-1 | **BostonU Club Tennis** (Community Service Chair): organized youth field day service event